Regular Expression

Basic Idea

- regular expression: defines a set of one or more strings of characters
- simple string of characters -> represents itself
- special/metacharacters -> characters that do not represent themselves
- add in special characters -> match a pattern which can represent many strings

Special Characters

- We'll focus on ERE (extended regular expression)
- delimiter:
  - marks beginning/end of expression,
  - often /
  - some utilities let you use other delimiters

Char	Use	Example
\	escape special character	a+b matches "a+b"
.	wildcard -match any character	.ord matches "word", "cord"
[ ]	character class	[bB]ob matches "bob", Bob"
^	beginning of line	^B matches "B" at start of line
$	end of line	!$ matches "!" ending line
*	match 0 or more occurrences	bo* matches "b", "boooo"
?	match 0 or 1 occurrences	bo+ matches "b", "bo"
+	match 1 or ore occurrences	bo+ matches "bo", "boooo"
{n}	match exactly n occurrences	bo{2} matches "boo"
{n, m}	match between n and m occurrences	bo{1,2} matches "bo", "boo"
()	group characters	(da)* matches "da", "dada"
|	match previous or next	hi|bye match "hi", "bye"